

Capstone Project

Customer Satisfaction Score Prediction

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Problem Statement

Objective :

- The goal is to accurately predict Customer Satisfaction (CSAT) scores using features derived from customer interactions.
- By predicting CSAT scores, we aim to proactively improve customer service and enhance overall satisfaction.

Challenges :

- Understanding key drivers of customer satisfaction.
- Identifying areas where service improvement can lead to higher customer satisfaction.
- Deploying the model for real-time predictions to support customer service operations.

Getting to know data

Dataset : (85907, 20)

Dataset Overview:

- **The dataset includes various features related to customer interactions, such as:**
- **CSAT Score:** The target variable representing customer satisfaction.
- **Agent Name, Issue Reported At, Issue Responded:** Key features reflecting the service process.
- **Customer Remarks:** Textual feedback provided by customers.
- **Product and Service Details:** Includes Product Category, Item Price, Channel Name, etc.

Getting to know data

- High null percentage is seen in columns like `order_date_time`, `customer_city`, `product_category`, `item_price`, `connected_handling_time` and `customer_reviews`.
- Other than customer remarks we will drop other columns as they are not useful.
- `Channel_name`, `category`, `sub-category`, `agent_shift` are some categorical columns
- `CSAT_Score` is ranges from 1 to 5

	Data type	#Missing	#Duplicate	#Unique
unique_id	object	0	0	85907
channel_name	object	0	0	3
category	object	0	0	12
sub-category	object	0	0	57
customer_remarks	object	57165	0	18231
order_id	object	18232	0	67675
order_date_time	object	68693	0	13766
issue_reported_at	object	0	0	30923
issue_responded	object	0	0	30262
survey_response_date	object	0	0	31
customer_city	object	68828	0	1782
product_category	object	68711	0	9
item_price	float64	68701	0	2789
connected_handling_time	float64	85665	0	211
agent_name	object	0	0	1371
supervisor	object	0	0	40
manager	object	0	0	6
tenure_bucket	object	0	0	5
agent_shift	object	0	0	5
csat_score	int64	0	0	5

Data Preprocessing & Feature Engineering

Data Cleaning:

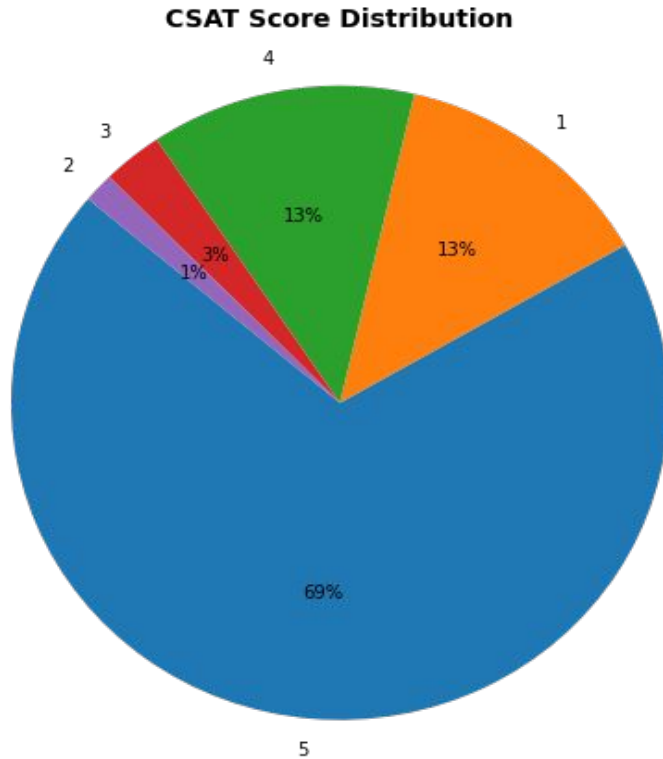
- Missing values were addressed, particularly in customer remarks where we filled missing entries with "Good" based on mode analysis.
- Date and time columns were converted to the appropriate datetime format.

Feature Engineering:

- Wait Response Time: Calculated as the difference between Issue Reported At and Issue Responded.
- Sentiment Analysis: Extracted sentiment scores from customer remarks to quantify customer feedback using VADER compound score. This score ranges from -1 to 1.
- Dropped unuseful columns like order_id, unique_id, columns with high nulls, date columns and customer reviews and checked for duplicates.

Prepared a clean, consistent dataset ready for further analysis and modeling.

CSAT Rating Distribution



- 70% of customer calls received a perfect rating of 5, indicating strong overall satisfaction. However, 13% of calls received a rating of 1, reflecting severe dissatisfaction.
- Focus on minimizing the low ratings (1s) to elevate customer service from "Excellent" to "Outstanding."

CSAT Score Distribution Sub-Category

Categories where CSAT score 5 exceeds 80%:

- - Billing Related: 80.70%
- - Instant discount: 80.77%
- - Issues with Shopzilla App: 81.08%
- - Missing: 81.26%
- - Non Order related: 100.00%
- - Payment pending: 88.24%
- - Signup Issues: 84.05%

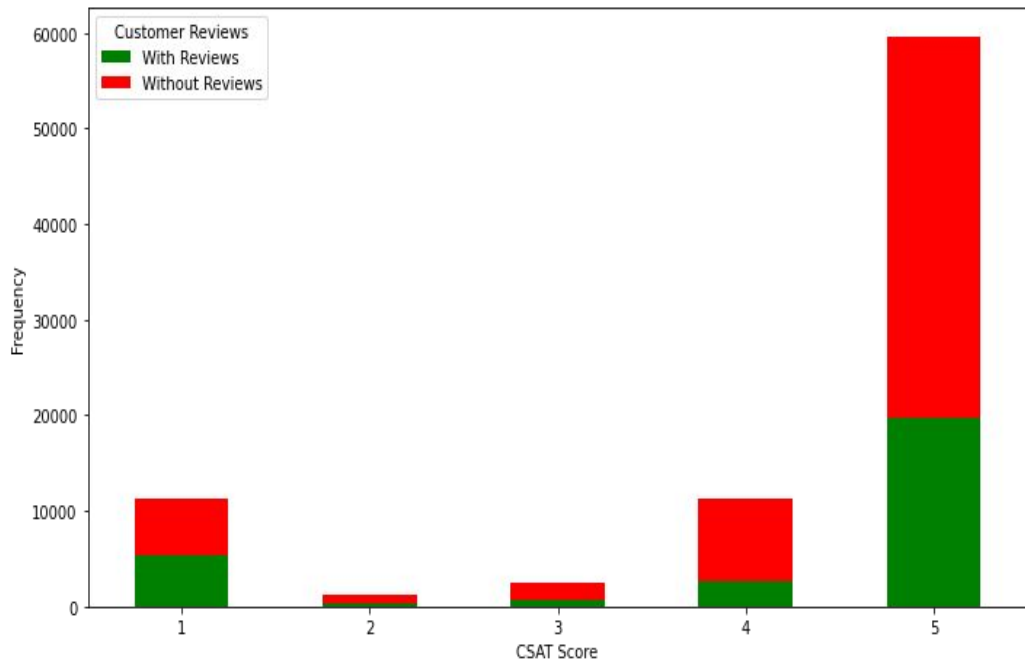
Categories where CSAT score 1 exceeds 25%:

- - Call disconnected: 35.00%
- - Card/EMI: 26.32%
- - Commission related: 66.67%
- -Other Account Related Issues: 31.82%
- - Others: 33.33%
- - Seller Cancelled Order: 28.80%
- - Service Center - Service Denial: 36.21%
- - Technician Visit: 32.80%
- - Unable to Login: 57.14%

Target these low-performing categories with specific interventions to reduce dissatisfaction.

Customer Remarks and CSAT Scores

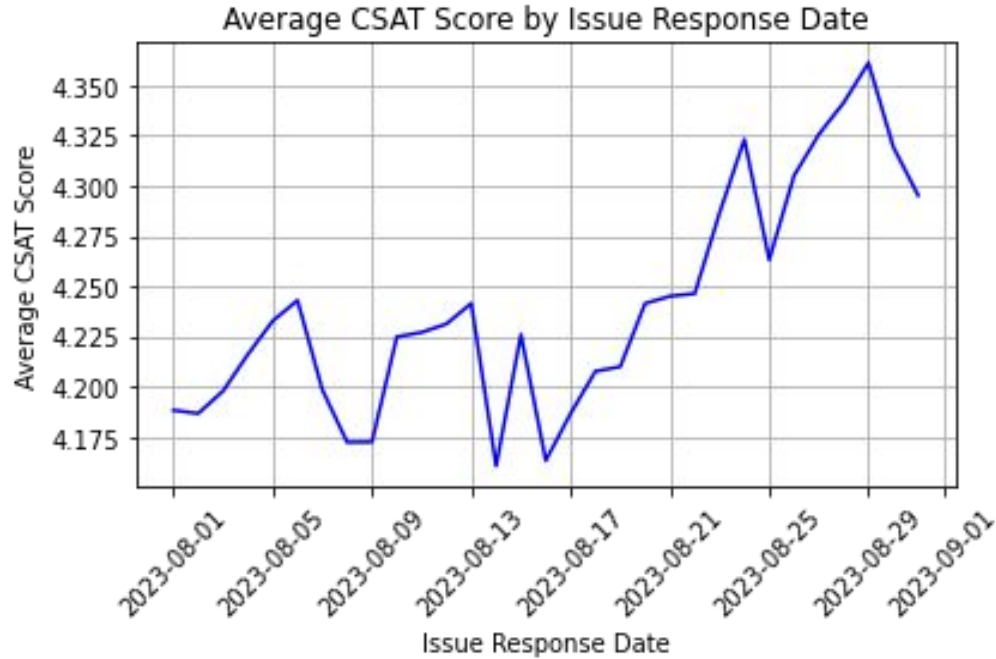
CSAT Score Distribution (With vs. Without Reviews)



- It becomes evident from this graph that Customer remarks are often provided when customers are either highly satisfied or highly disappointed.
- So the mode “Good” is imputed if customer review is absent.

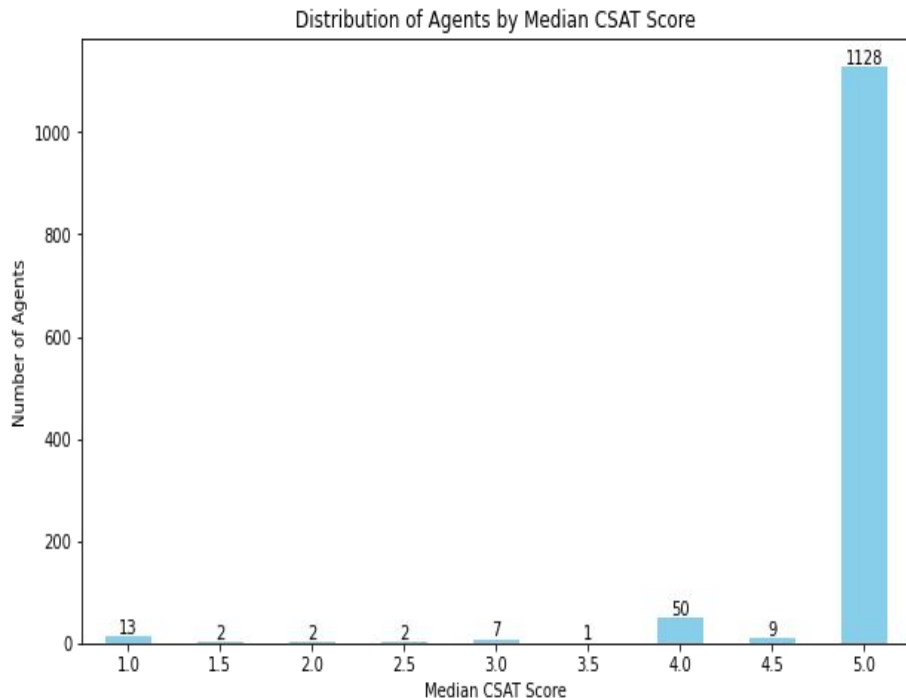
	With Reviews	Without Reviews
csat_score		
1	5348	5880
2	394	889
3	730	1828
4	2603	8615
5	19667	39944

Trend in CSAT Scores



- An upward trend in CSAT scores was observed, indicating improving customer satisfaction over time.
- Monitoring and maintain this positive trend by focusing on areas with persistent low satisfaction gives a positive impact.

Focus on Low-Performing Agents



- This graph illustrates the performance distribution among agents, revealing that 1,187 agents have a median CSAT score of 4 or higher. In contrast, 27 agents have a median score below 4.
- This indicates a need to focus on these lower-performing agents, either through targeted training or by investigating the specific issues they consistently encounter to drive improvement.

Data Transformations & Normalization



Scaling:

- Features (X) and the target variable (y) were normalized using MinMaxScaler to bring all variables into a consistent range, facilitating smoother training and improved model performance.

Encoding:

- One-Hot Encoding: Applied to categorical variables like Channel Name, Category, Tenure Bucket, and Agent Shift to convert them into numerical format.
- Target Encoding: Used for Sub-Category, Agent Name, Supervisor, and Manager to capture their relationship with CSAT scores.

Therefore the dataset was fully transformed and ready for modeling. These encodings are saved in pkl files for predictions pipeline, ensuring that the same transformations were applied consistently across training and test datasets.

Modelling - Architecture

Neural Network Design:

1. **Layers:** The model is a deep neural network with two hidden layers. Each hidden layer has 64 neurons and uses the ReLU activation function, which helps the network learn complex patterns by introducing non-linearity.
2. **Dropout Regularization:** Dropout is applied after each hidden layer with a dropout rate of 20%. This technique randomly drops neurons during training to prevent overfitting, ensuring that the model generalizes well to new data.
3. **Output Layer:** The output layer consists of a single neuron with a linear activation function, suitable for predicting continuous CSAT scores.

Modelling - Architecture

- **Optimizers and Learning Rate:**
 1. **Adam Optimizer:**The Adam optimizer was chosen for its adaptive learning rate capabilities and efficiency in handling sparse gradients.
 2. **Decaying Learning Rate:**An exponentially decaying learning rate was implemented to gradually reduce the step size as the model trains. This helps in fine-tuning the model, allowing it to converge more smoothly and avoid overshooting the optimal point.
- **Callbacks:**
 1. **Early Stopping:**An early stopping callback was used to monitor the validation loss. Training is halted when there is no improvement for 5 consecutive epochs, which prevents overfitting and saves computational resources.
 2. **Learning Rate Scheduler:**A learning rate scheduler was used to adjust the learning rate dynamically during training, based on the performance of the model.

Modelling - Hyperparameter tuning

RandomizedSearchCV: A RandomizedSearchCV was used to optimize key hyperparameters, including the number of neurons, dropout rate, batch size, epochs. This approach randomly samples from a specified range of values, allowing for a more efficient search over a large hyperparameter space compared to grid search.

- **Key Parameters Tuned:**
 1. **Neurons:** Number of neurons in each hidden layer, 32 to 128.
 2. **Dropout Rate:** Proportion of neurons dropped during training, 10 to 40%
 3. **Batch Size:** Number of samples processed before the model's internal parameters are updated, ranging from 16 to 64.
 4. **Epochs:** Total number of iterations over the entire dataset during training, 10 to 150.

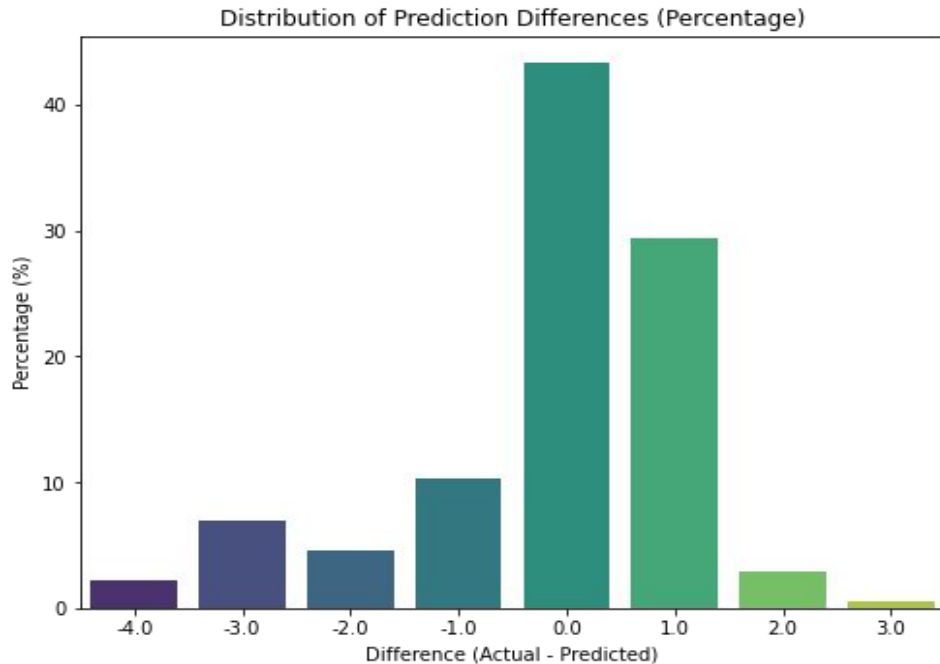
The RandomizedSearchCV identified the best combination of hyperparameters that led to the most accurate and efficient model. The optimized model was able to achieve better generalization with reduced overfitting.

Modelling - Evaluation



Achieved ~70% of accuracy on both training and validation ensuring there is no sign of overfitting.

Modelling - Evaluation

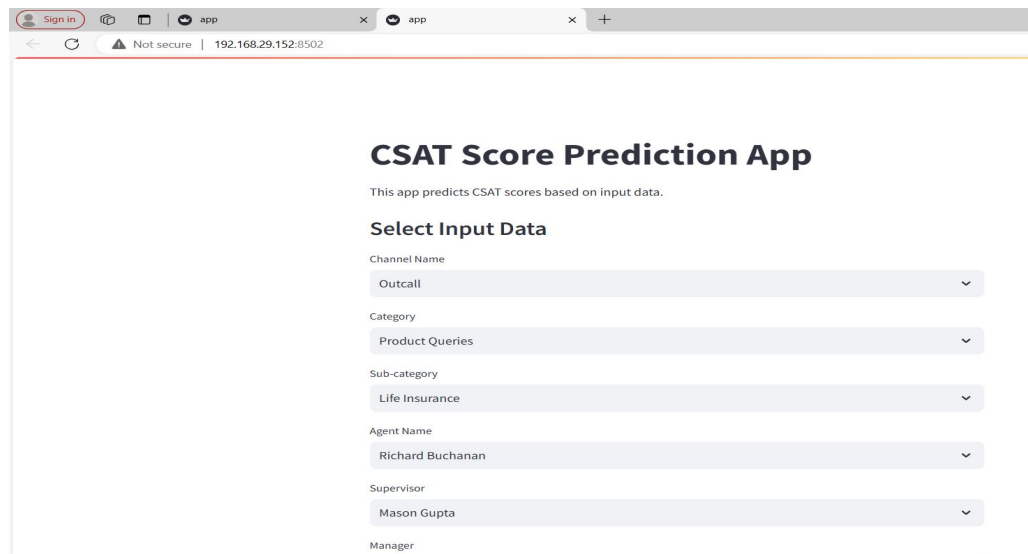


This plot will visually represent how frequently the model's predictions differ from the actual values by specific amounts, expressed as a percentage of the total predictions. This is helpful to understand the model's accuracy and error distribution.

Deployment

Local URL: <http://localhost:8502>

Network URL: <http://192.168.29.152:8502>



The screenshot shows a web browser window with the address bar displaying "Not secure | 192.168.29.152:8502". The page title is "CSAT Score Prediction App". Below the title, a subtitle reads "This app predicts CSAT scores based on input data." The main section is titled "Select Input Data" and contains six dropdown menus for input fields: "Channel Name" (selected: Outcall), "Category" (selected: Product Queries), "Sub-category" (selected: Life Insurance), "Agent Name" (selected: Richard Buchanan), "Supervisor" (selected: Mason Gupta), and "Manager" (empty).

CSAT Score Prediction App

This app predicts CSAT scores based on input data.

Select Input Data

Channel Name
Outcall

Category
Product Queries

Sub-category
Life Insurance

Agent Name
Richard Buchanan

Supervisor
Mason Gupta

Manager

Action Points

- 1. Optimize Customer Service Operations:** Reduce response times, as they significantly impact customer satisfaction. Consider increasing staffing during peak periods and implementing automation where possible.
- 2. Improve Low-Performing Agents:** Provide targeted training and support for the 27 agents with a median CSAT score below 4. Conduct detailed performance reviews to address specific challenges.
- 3. Enhance Data Quality and Model Accuracy:** Continuously update the dataset and explore additional features that could improve the model's predictive power. Focus on accurate predictions, particularly for cases with larger discrepancies.
- 4. Proactive Customer Engagement:** Use the upward trend in CSAT scores to explore new ways to exceed customer expectations, particularly in high-performing areas like billing and app-related issues.

Conclusion

- The CSAT prediction model has successfully identified key drivers of customer satisfaction, allowing for targeted improvements in service quality.
- The model's predictions can be used to proactively address potential issues, ensuring higher levels of customer satisfaction.
- The insights provided offer a clear path to enhancing customer service, improving agent performance, and ultimately increasing customer loyalty.

Q & A